Pearson Edexcel A-Level Mathematics (9MA0)

Module 5

Applied: Statistics & Mechanics

Paper 3 · 69 practice questions · answers and mark schemes at the back

Subtopic	Questions	Done
5.1 Statistics	36	____ / 36
5.2 Mechanics	33	____ / 33

How to use this workbook

1. Read the key facts and watch one video for the subtopic. Tick the skills checklist as you go.
2. Attempt each question in the squared working space without notes. Mark it with the scheme at the back.
3. Tick **Todo** when you have attempted it and **Mastered** only when you can redo it cold a week later.
4. Copy any mistakes into the notes box at the end of the subtopic. Rebuild the workbook with a new seed for fresh questions.

Name _____ Started _____ Built 14 Oct 2026, seed 2026

5.1 Statistics

Statistics (Paper 3 Section A): sampling, data presentation, the large data set, probability, binomial and normal distributions, regression/correlation and hypothesis testing.

Key facts

- $X \sim \mathrm{B}(n, p)$: $\mathrm{P}(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$, mean np, variance $np(1-p)$
- $X \sim \mathrm{N}(\mu, \sigma^2)$: $Z = \frac{X-\mu}{\sigma} \sim \mathrm{N}(0, 1)$
- Normal approximation to $\mathrm{B}(n, p)$: $\mathrm{N}(np, np(1-p))$ with continuity correction
- $\bar{X} \sim \mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right)$ for a sample mean
- $\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$; independent iff $\mathrm{P}(A \cap B) = \mathrm{P}(A)\mathrm{P}(B)$
- $\sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2$; outlier rule: beyond $Q_1 - 1.5\,\mathrm{IQR}$ or $Q_3 + 1.5\,\mathrm{IQR}$

Method

- Hypothesis test template: hypotheses (parameter!), distribution under H_0, probability or critical region, compare, conclusion in context.
- Critical region: find the largest/smallest value whose tail probability is below the significance level.
- Histograms: frequency density = frequency ÷ class width; area represents frequency.

Exam traps

- Writing hypotheses with $\bar{x}$ or X instead of p, μ or ρ.
- Conclusions with no context or that 'prove' something.
- Forgetting the continuity correction.
- Not knowing the large data set (stations, variables, 'tr', 'n/a').

In the exam: Paper 3 Section A is 50 marks: around 5 questions mixing the topics above.

Specification checklist

Skill	Learnt	Practised	Mastered
Populations and sampling methods	☐	☐	☐
Measures of location and spread, coding	☐	☐	☐
Histograms, box plots, cumulative frequency, outliers	☐	☐	☐
Correlation and regression	☐	☐	☐
The large data set	☐	☐	☐
Probability: Venn diagrams and set notation	☐	☐	☐
Conditional probability and tree diagrams	☐	☐	☐
Binomial distribution	☐	☐	☐
Hypothesis testing: critical regions and p-values	☐	☐	☐
Normal distribution	☐	☐	☐
Normal approximation to binomial	☐	☐	☐
Hypothesis testing: normal mean and correlation	☐	☐	☐

Videos (click to open)

☐ **Recap** — Probability (Year 1) in less than 20 minutes • A-Level Maths, Statistics Year 1, Chapter 5 (Bicen Maths)
☐ **Recap** — Statistical Distributions (+ binomial) in 29 minutes • A-Level Maths, Statistics Year 1, Chapter 6 (Bicen Maths)
☐ **Recap** — Hypothesis Testing (Binomial) in 24 minutes • A-Level Maths, Statistics Year 1, Chapter 7 (Bicen Maths)
☐ Populations and sampling methods — Sampling: Population vs Sample (TLMaths)
☐ Measures of location and spread, coding — Data: Introducing the Variance and Standard Deviation (TLMaths)
☐ Histograms, box plots, cumulative frequency, outliers — Data: Histograms (TLMaths)
☐ Correlation and regression — Scatter Graphs: The Product Moment Correlation Coefficient (TLMaths)
☐ The large data set — Edexcel Large Data Set - Weather (TLMaths)
☐ Probability: Venn diagrams and set notation — Probability: Venn Diagrams - Union and Intersection (TLMaths)
☐ Conditional probability and tree diagrams — Probability: Introducing Conditional Probability (TLMaths)
☐ Binomial distribution — Binomial Distribution: Introducing the Binomial Distribution (TLMaths)
☐ Hypothesis testing: critical regions and p-values — Binomial Hypothesis Testing: Finding the Critical Region (TLMaths)

☐ Normal distribution — Normal Distribution: Introducing the Bell Curve (TLMaths)
☐ Normal approximation to binomial — Normal Distribution: Approximating a Binomial Distribution (TLMaths)
☐ Hypothesis testing: normal mean and correlation — Introducing Sample Means Hypothesis Testing (TLMaths)

Q5.1.01 Histograms, box plots, cumulative frequency, outliers ●○○ [4 marks] ☐ Todo ☐ Mastered

The times, t minutes, taken by some students are grouped as: $0 \le t < 10$: 5, $10 \le t < 20$: 13, $20 \le t < 30$: 16, $30 \le t < 50$: 22, $50 \le t < 80$: 13.

(a) Find the frequency density of each class for a histogram. **(2)**

(b) For a different data set, $Q_1 = 29$ and $Q_3 = 54$. Outliers are values more than $1.5 \times$ IQR beyond the quartiles. Find the outlier boundaries. **(2)**

WORKING

Q5.1.02 Measures of location and spread, coding ●○○ [5 marks] ☐ Todo ☐ Mastered

For 17 values of x: $\sum x = 635$ and $\sum x^2 = 27329$.

(a) Find the mean and standard deviation of x. **(3)**

(b) The data are coded using $y = \frac{x-5}{10}$. Find the mean and standard deviation of y. **(2)**

WORKING

Q5.1.03 The large data set ●○○ [2 marks] ☐ Todo ☐ Mastered

Explain why data from the large data set for Perth (Australia) in May might not be suitable for estimating UK summer rainfall.

(2)

WORKING

Q5.1.04 Binomial distribution ●○○ [7 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{B}(10, 0.35)$. Find

(a) $\mathrm{P}(X = 3)$, (b) $\mathrm{P}(X \leq 6)$, (c) $\mathrm{P}(X > 3)$, (d) $\mathrm{P}(3 < X \leq 6)$.

(a) **(2)**

(b) **(1)**

(c) **(2)**

(d) **(2)**

WORKING

Q5.1.05 Probability: Venn diagrams and set notation ●○○ [9 marks] ☐ Todo ☐ Mastered

In a group of 50 students, 19 study Art, 13 study Biology and 26 study neither.

(a) Draw a Venn diagram. **(3)**

(b) Find $\mathrm{P}(A \cap B)$, $\mathrm{P}(A \cup B')$ and $\mathrm{P}(A \mid B)$. **(4)**

(c) Determine whether A and B are independent. **(2)**

WORKING

Q5.1.06 Populations and sampling methods ●○○ [2 marks] ☐ Todo ☐ Mastered

Give one advantage and one disadvantage of quota sampling compared with stratified sampling.

(2)

WORKING

Q5.1.07 Correlation and regression ●○○ [3 marks] ☐ Todo ☐ Mastered

The regression line of y, ice creams sold, on x, daily mean temperature (°C), is $y = 15.4 + 0.31x$.

(a) Interpret the gradient in context. **(1)**

(b) Estimate y when $x = 26$ and comment on reliability if the data ranged from 2 to 20. **(2)**

WORKING

Q5.1.08 Histograms, box plots, cumulative frequency, outliers ●○○ [4 marks] ☐ Todo ☐ Mastered

The times, t minutes, taken by some students are grouped as: $0 \le t < 10$: 14, $10 \le t < 20$: 11, $20 \le t < 30$: 29, $30 \le t < 50$: 11, $50 \le t < 80$: 26.

(a) Find the frequency density of each class for a histogram. **(2)**

(b) For a different data set, $Q_1 = 23$ and $Q_3 = 41$. Outliers are values more than $1.5 \times$ IQR beyond the quartiles. Find the outlier boundaries. **(2)**

WORKING

Q5.1.09 Measures of location and spread, coding ●○○ [5 marks] ☐ Todo ☐ Mastered

For 23 values of x: $\sum x = 828$ and $\sum x^2 = 35852$.

(a) Find the mean and standard deviation of x. **(3)**

(b) The data are coded using $y = \frac{x - 8}{10}$. Find the mean and standard deviation of y. **(2)**

WORKING

Q5.1.10 The large data set ●○○ [2 marks] ☐ Todo ☐ Mastered

The large data set records 'tr' for daily total rainfall. State what this means and the value used in calculations.

(2)

WORKING

Q5.1.11 Binomial distribution ●○○ [7 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{B}(14, 0.25)$. Find

(a) $\mathrm{P}(X = 3)$, (b) $\mathrm{P}(X \leq 3)$, (c) $\mathrm{P}(X > 3)$, (d) $\mathrm{P}(3 < X \leq 3)$.

(a) **(2)**

(b) **(1)**

(c) **(2)**

(d) **(2)**

WORKING

Q5.1.12 Probability: Venn diagrams and set notation ●○○ [9 marks] ☐ Todo ☐ Mastered

In a group of 60 students, 20 study Art, 19 study Biology and 28 study neither.

(a) Draw a Venn diagram. **(3)**

(b) Find $P(A \cap B)$, $P(A \cup B')$ and $P(A \mid B)$. **(4)**

(c) Determine whether A and B are independent. **(2)**

WORKING

Q5.1.13 Populations and sampling methods ●○○ [3 marks] ☐ Todo ☐ Mastered

Describe how to take a simple random sample of 40 students from a school of 900.

(3)

WORKING

Q5.1.14 Correlation and regression ●○○ [3 marks] ☐ Todo ☐ Mastered

The regression line of y, ice creams sold, on x, daily mean temperature (°C), is $y = 30.1 + 3.92x$.

(a) Interpret the gradient in context. **(1)**

(b) Estimate y when $x = 17$ and comment on reliability if the data ranged from 2 to 20. **(2)**

WORKING

Q5.1.15 Histograms, box plots, cumulative frequency, outliers ●○○ [4 marks] ☐ Todo ☐ Mastered

The times, t minutes, taken by some students are grouped as: $0 \le t < 10$: 12, $10 \le t < 20$: 8, $20 \le t < 30$: 25, $30 \le t < 50$: 13, $50 \le t < 80$: 28.

(a) Find the frequency density of each class for a histogram. **(2)**

(b) For a different data set, $Q_1 = 26$ and $Q_3 = 45$. Outliers are values more than $1.5 \times$ IQR beyond the quartiles. Find the outlier boundaries. **(2)**

WORKING

Q5.1.16 Measures of location and spread, coding ●○○ [5 marks] ☐ Todo ☐ Mastered

For 29 values of x: $\sum x = 1063$ and $\sum x^2 = 45661$.

(a) Find the mean and standard deviation of x. **(3)**

(b) The data are coded using $y = \frac{x - 10}{5}$. Find the mean and standard deviation of y. **(2)**

WORKING

Q5.1.17 The large data set ●○○ [2 marks] ☐ Todo ☐ Mastered

Using your knowledge of the large data set, explain why daily mean windspeed for Leeming may contain the entry 'n/a' and how you would clean such data.

(2)

WORKING

Q5.1.18 Binomial distribution ●○○ [7 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{B}(24, 0.5)$. Find

(a) $\mathrm{P}(X = 5)$, (b) $\mathrm{P}(X \leq 17)$, (c) $\mathrm{P}(X > 5)$, (d) $\mathrm{P}(5 < X \leq 17)$.

(a) **(2)**

(b) **(1)**

(c) **(2)**

(d) **(2)**

WORKING

Q5.1.19 Probability: Venn diagrams and set notation ●○○ [9 marks] ☐ Todo ☐ Mastered

In a group of 60 students, 23 study Art, 22 study Biology and 24 study neither.

(a) Draw a Venn diagram. **(3)**

(b) Find $\mathrm{P}(A \cap B)$, $\mathrm{P}(A \cup B')$ and $\mathrm{P}(A \mid B)$. **(4)**

(c) Determine whether A and B are independent. **(2)**

WORKING

Q5.1.20 Populations and sampling methods ●○○ [2 marks] ☐ Todo ☐ Mastered

Explain the difference between a census and a sample, giving one advantage of each.

(2)

WORKING

Q5.1.21 Correlation and regression ●○○ [3 marks] ☐ Todo ☐ Mastered

The regression line of y, test score, on x, hours of revision, is $y = 13.1 + 3.36x$.

(a) Interpret the gradient in context. **(1)**

(b) Estimate y when $x = 23$ and comment on reliability if the data ranged from 2 to 20. **(2)**

WORKING

Q5.1.22 Normal approximation to binomial ●●○ [5 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{B}(177, 0.4)$.

(a) Explain why a normal approximation is suitable. **(1)**

(b) Use it to estimate $\mathrm{P}(X \leq 72)$. **(4)**

WORKING

Q5.1.23 Hypothesis testing: critical regions and p-values ●●○ [5 marks] ☐ Todo ☐ Mastered

Historically, a seed company claims that 40% of its seeds produce red flowers. After a change, a random sample of 40 is taken and 13 are found to have the property. Test, at the 10% level, whether there is evidence that the proportion has decreased. State your hypotheses clearly.

(5)

WORKING

Q5.1.24 Conditional probability and tree diagrams ●●○ [6 marks] ☐ Todo ☐ Mastered

The probability that a person has a condition is $\frac{1}{3}$. If so, the probability that a person tests positive is $\frac{4}{5}$; otherwise it is $\frac{1}{5}$.

(a) Find the probability that a person tests positive. **(3)**

(b) Given that a person tests positive, find the probability that the person has the condition. **(3)**

WORKING

Q5.1.25 Normal distribution ●●○ [3 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{N}(88, 4^2)$. Find the value of a such that $\mathrm{P}(X < a) = 0.8$.

(3)

WORKING

Q5.1.26 Normal approximation to binomial ●●○ [5 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{B}(62, 0.5)$.

(a) Explain why a normal approximation is suitable. **(1)**

(b) Use it to estimate $\mathrm{P}(X \leq 36)$. **(4)**

WORKING

Q5.1.27 **Hypothesis testing: critical regions and p-values** ●●○ **[5 marks]** ☐ **Todo** ☐ **Mastered**

Historically, 50% of patients normally recover within a week. After a change, a random sample of 20 is taken and 5 are found to have the property. Test, at the 5% level, whether there is evidence that the proportion has decreased. State your hypotheses clearly.

(5)

WORKING

Q5.1.28 **Conditional probability and tree diagrams** ●●○ **[6 marks]** ☐ **Todo** ☐ **Mastered**

The probability that a person has a condition is $\frac{2}{5}$. If so, the probability that a person tests positive is $\frac{2}{3}$; otherwise it is $\frac{1}{4}$.

(a) Find the probability that a person tests positive. **(3)**

(b) Given that a person tests positive, find the probability that the person has the condition. **(3)**

WORKING

Q5.1.29 Normal approximation to binomial ●●○ [5 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{B}(65, 0.55)$.

(a) Explain why a normal approximation is suitable. **(1)**

(b) Use it to estimate $\mathrm{P}(X \leq 31)$. **(4)**

WORKING

Q5.1.30 Hypothesis testing: critical regions and p-values ●●○ [5 marks] ☐ Todo ☐ Mastered

Historically, a website claims that 20% of visitors make a purchase. After a change, a random sample of 20 is taken and 6 are found to have the property. Test, at the 10% level, whether there is evidence that the proportion has increased. State your hypotheses clearly.

(5)

WORKING

Q5.1.31 Conditional probability and tree diagrams ●●○ [6 marks] ☐ Todo ☐ Mastered

The probability that it rains on a given day is $\frac{3}{10}$. If so, the probability that Ben is late for school is $\frac{4}{5}$; otherwise it is $\frac{1}{10}$.

(a) Find the probability that Ben is late for school. **(3)**

(b) Given that Ben is late for school, find the probability that it rained. **(3)**

WORKING

Q5.1.32 Normal distribution ●●● [6 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{N}(\mu, \sigma^2)$. Given $\mathrm{P}(X < 114.4) = 0.2$ and $\mathrm{P}(X > 133.5) = 0.1$, find μ and σ.

(6)

WORKING

Q5.1.33 Hypothesis testing: normal mean and correlation ●●● [6 marks] ☐ Todo ☐ Mastered

The masses of items are normally distributed with standard deviation 17 g. The mean is claimed to be 90 g. A random sample of 36 items has mean 86.9 g. Test, at the 1% level, whether the mean is less than 90 g.

(6)

WORKING

Q5.1.34 Hypothesis testing: normal mean and correlation ●●● [6 marks] ☐ Todo ☐ Mastered

The masses of items are normally distributed with standard deviation 15 g. The mean is claimed to be 54 g. A random sample of 20 items has mean 46.0 g. Test, at the 5% level, whether the mean is less than 54 g.

(6)

WORKING

Q5.1.35 Normal distribution ●●● [6 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{N}(\mu, \sigma^2)$. Given $\mathrm{P}(X < 120.5) = 0.1$ and $\mathrm{P}(X > 139.6) = 0.2$, find μ and σ.

(6)

WORKING

Q5.1.36 Hypothesis testing: normal mean and correlation ●●● [6 marks] ☐ Todo ☐ Mastered

The masses of items are normally distributed with standard deviation 9 g. The mean is claimed to be 125 g. A random sample of 20 items has mean 128.3 g. Test, at the 5% level, whether the mean is greater than 125 g.

(6)

WORKING

MY NOTES — 5.1 STATISTICS: MISTAKES TO REMEMBER

5.2 Mechanics

Mechanics (Paper 3 Section B): modelling, kinematics (suvat, graphs, calculus, vectors), forces and Newton's laws, friction, connected particles, moments and projectiles.

Key facts

- $v = u + at,\ \ s = ut + \frac{1}{2}at^2,\ \ s = vt - \frac{1}{2}at^2,\ \ v^2 = u^2 + 2as,\ \ s = \frac{1}{2}(u+v)t$
- $v = \frac{dr}{dt},\ \ a = \frac{dv}{dt};\ \ r = \int v\,dt,\ \ v = \int a\,dt$
- $\mathbf{F} = m\mathbf{a}$; weight $= mg$ with $g = 9.8$ m s^{-2}
- Friction $F \le \mu R$; limiting equilibrium/moving: $F = \mu R$
- Moment = force $\times$ perpendicular distance; equilibrium: resultant force 0 and total moment 0
- Projectiles: horizontal $x = U\cos\alpha\, t$; vertical $y = U\sin\alpha\, t - \frac{1}{2}gt^2$

Method

- Draw a force diagram for every body; resolve parallel and perpendicular to the motion.
- Connected particles: write $F = ma$ for each particle separately (same a, same T for a light inextensible string).
- Moments: take moments about the point with the most unknown forces.

Exam traps

- Using $g = 9.81$ or 10 instead of 9.8.
- Giving too many significant figures after using $g = 9.8$ (2 or 3 s.f.).
- Forgetting that friction opposes the direction of (potential) motion.

In the exam: Paper 3 Section B is 50 marks; projectiles, connected particles and moments are each worth 8-12 marks.

Specification checklist

Skill	Learnt	Practised	Mastered
Quantities and units, modelling assumptions	☐	☐	☐
Constant acceleration: suvat	☐	☐	☐
Motion graphs	☐	☐	☐
Variable acceleration with calculus	☐	☐	☐
2D motion with vectors	☐	☐	☐
Newton's laws and resolving forces	☐	☐	☐
Friction	☐	☐	☐
Connected particles and pulleys	☐	☐	☐
Inclined planes	☐	☐	☐
Moments, equilibrium and centre of mass	☐	☐	☐
Projectiles	☐	☐	☐

Videos (click to open)

☐ **Recap** — Constant Acceleration (SUVAT) in 31 minutes • A-Level Maths, Mechanics Year 1, Chapter 9 (Bicen Maths)
☐ **Recap** — Variable Acceleration in less than 12 minutes • A-Level Maths, Mechanics Year 1, Chapter 11 (Bicen Maths)
☐ **Recap** — Forces & Motion (Year 1) in less than 37 minutes • A-Level Maths, Mechanics Year 1, Chapter 10 (Bicen Maths)
☐ Quantities and units, modelling assumptions — Introduction to Modelling (TLMaths)
☐ Constant acceleration: suvat — SUVAT: Deriving the Constant Acceleration Formulae (TLMaths)
☐ Motion graphs — Kinematics: Velocity / Time Graphs (TLMaths)
☐ Variable acceleration with calculus — variable acceleration - calculus methods Example (1) (ExamSolutions)
☐ 2D motion with vectors — Vectors: Kinematics Problem (TLMaths)
☐ Newton's laws and resolving forces — Forces: Resolving Forces Example 1 (TLMaths)
☐ Friction — Forces: Introducing the Coefficient of Friction (TLMaths)
☐ Connected particles and pulleys — Forces: Pulley Example 1 (TLMaths)
☐ Inclined planes — Forces: Introducing Smooth Inclined Planes (TLMaths)
☐ Moments, equilibrium and centre of mass — Moments: What is a Moment? (TLMaths)
☐ Projectiles — Projectiles: Introducing Projectiles (TLMaths)

Q5.2.01 Friction ●○○ [4 marks] ☐ Todo ☐ Mastered

A box of mass 8 kg on rough horizontal ground ($\mu = 0.3$) is pulled by a horizontal force of 41 N. Find its acceleration.

(4)

WORKING

Q5.2.02 Quantities and units, modelling assumptions ●○○ [2 marks] ☐ Todo ☐ Mastered

In mechanics, explain the meaning of the modelling term 'particle' and its effect on calculations.

(2)

WORKING

Q5.2.03 Motion graphs ●○○ [5 marks] ☐ Todo ☐ Mastered

A train starts from rest and accelerates uniformly for 8 s to 12 m s^{-1}, travels at this speed for 17 s, then decelerates uniformly to rest in 5 s.

(a) Sketch a velocity-time graph. **(2)**

(b) Find the total distance travelled. **(2)**

(c) Find the deceleration. **(1)**

WORKING

Q5.2.04 Friction ●○○ [4 marks] ☐ Todo ☐ Mastered

A box of mass 3 kg on rough horizontal ground ($\mu = 0.4$) is pulled by a horizontal force of 71 N. Find its acceleration.

(4)

WORKING

Q5.2.05 Quantities and units, modelling assumptions ●○○ [2 marks] ☐ Todo ☐ Mastered

In mechanics, explain the meaning of the modelling term 'rigid' and its effect on calculations.

(2)

WORKING

Q5.2.06 Motion graphs ●○○ [5 marks] ☐ Todo ☐ Mastered

A train starts from rest and accelerates uniformly for 8 s to 9 m s^{-1}, travels at this speed for 25 s, then decelerates uniformly to rest in 2 s.

(a) Sketch a velocity-time graph. **(2)**

(b) Find the total distance travelled. **(2)**

(c) Find the deceleration. **(1)**

WORKING

Q5.2.07 Friction ●○○ [4 marks] ☐ Todo ☐ Mastered

A box of mass 5 kg on rough horizontal ground ($\mu = 0.1$) is pulled by a horizontal force of 40 N. Find its acceleration.

(4)

WORKING

Q5.2.08 Quantities and units, modelling assumptions ●○○ [2 marks] ☐ Todo ☐ Mastered

In mechanics, explain the meaning of the modelling term 'uniform rod' and its effect on calculations.

(2)

WORKING

Q5.2.09 Motion graphs ●○○ [5 marks] ☐ Todo ☐ Mastered

A train starts from rest and accelerates uniformly for 5 s to 18 m s^{-1}, travels at this speed for 21 s, then decelerates uniformly to rest in 8 s.

(a) Sketch a velocity-time graph. **(2)**

(b) Find the total distance travelled. **(2)**

(c) Find the deceleration. **(1)**

WORKING

Q5.2.10 Connected particles and pulleys ●●○ [8 marks] ☐ Todo ☐ Mastered

Particles A (2 kg) and B (6 kg) are connected by a light inextensible string over a smooth fixed pulley. The system is released from rest with B 4 m above the ground.

(a) Find the acceleration of the system and the tension in the string. **(5)**

(b) Find the speed with which B hits the ground. **(2)**

(c) State how you used the fact that the string is inextensible. **(1)**

WORKING

Q5.2.11 Moments, equilibrium and centre of mass ●●○ [5 marks] ☐ Todo ☐ Mastered

A uniform beam AB of length 4 m and weight 80 N rests horizontally on supports at A and B. A load of 72 N is placed 2 m from A. Find the reactions at the supports, and state one modelling assumption.

(5)

WORKING

Q5.2.12 Newton's laws and resolving forces ●●○ [5 marks] ☐ Todo ☐ Mastered

A particle of weight 19 N hangs in equilibrium from two light strings inclined at 30° and 60° to the horizontal. Find the tension in each string.

(5)

WORKING

Q5.2.13 Inclined planes ●●○ [6 marks] ☐ Todo ☐ Mastered

A particle of mass 5 kg slides down a rough plane inclined at α to the horizontal, where $\tan\alpha = \frac{7}{24}$. The coefficient of friction is 0.1.

Find the normal reaction and the acceleration of the particle.

(6)

WORKING

Q5.2.14 Variable acceleration with calculus ●●○ [5 marks] ☐ Todo ☐ Mastered

A particle moves in a straight line with velocity $v = -t^2 - 3t + 4$ m s^{-1} at time t s.

(a) Find the acceleration when $t = 5$. **(2)**

(b) Find the displacement from the starting point when $t = 5$. **(3)**

WORKING

Q5.2.15 Projectiles ●●○ [7 marks] ☐ Todo ☐ Mastered

A particle is projected from a point O on horizontal ground with speed 23 m s^{-1} at angle α above the horizontal, where $\tan\alpha = \frac{5}{12}$.

Find (a) the time of flight, (b) the range, (c) the greatest height.

(a) **(3)**

(b) **(2)**

(c) **(2)**

WORKING

Q5.2.16 2D motion with vectors ●●○ [4 marks] ☐ Todo ☐ Mastered

A particle has initial velocity $(5\mathbf{i} - \mathbf{j})$ m s^{-1} and constant acceleration $(-2\mathbf{i} - \mathbf{j})$ m s^{-2}. Find its speed after 4 s and the angle its direction of motion makes with $\mathbf{i}$.

(4)

WORKING

Q5.2.17 Constant acceleration: suvat ●●○ [7 marks] ☐ Todo ☐ Mastered

A ball is projected vertically upwards at 15 m s^{-1} from a point 30 m above the ground.

(a) Find the greatest height above the ground. **(3)**

(b) Find the time taken to reach the ground. **(3)**

(c) State one modelling assumption. **(1)**

WORKING

Q5.2.18 Connected particles and pulleys ●●○ [8 marks] ☐ Todo ☐ Mastered

Particles A (3 kg) and B (5 kg) are connected by a light inextensible string over a smooth fixed pulley. The system is released from rest with B 3 m above the ground.

(a) Find the acceleration of the system and the tension in the string. **(5)**

(b) Find the speed with which B hits the ground. **(2)**

(c) State how you used the fact that the string is inextensible. **(1)**

WORKING

Q5.2.19 Moments, equilibrium and centre of mass ●●○ [5 marks] ☐ Todo ☐ Mastered

A uniform beam AB of length 5 m and weight 169 N rests horizontally on supports at A and B. A load of 52 N is placed 0.5 m from A. Find the reactions at the supports, and state one modelling assumption.

(5)

WORKING

Q5.2.20 Newton's laws and resolving forces ●●○ [5 marks] ☐ Todo ☐ Mastered

A particle of weight 42 N hangs in equilibrium from two light strings inclined at 40° and 50° to the horizontal. Find the tension in each string.

(5)

WORKING

Q5.2.21 Inclined planes ●●○ [6 marks] ☐ Todo ☐ Mastered

A particle of mass 5 kg slides down a rough plane inclined at α to the horizontal, where $\tan\alpha = \frac{5}{12}$. The coefficient of friction is 0.2.
Find the normal reaction and the acceleration of the particle.

(6)

WORKING

Q5.2.22 Variable acceleration with calculus ●●○ [5 marks] ☐ Todo ☐ Mastered

A particle moves in a straight line with velocity $v = -3t^2 - 10t + 5$ m s^{-1} at time t s.

(a) Find the acceleration when $t = 3$. **(2)**

(b) Find the displacement from the starting point when $t = 3$. **(3)**

WORKING

Q5.2.23 Projectiles ●●○ [7 marks] ☐ Todo ☐ Mastered

A particle is projected from a point O on horizontal ground with speed 16 m s^{-1} at angle α above the horizontal, where $\tan\alpha = \frac{4}{3}$.

Find (a) the time of flight, (b) the range, (c) the greatest height.

(a) **(3)**

(b) **(2)**

(c) **(2)**

WORKING

Q5.2.24 2D motion with vectors ●●○ [4 marks] ☐ Todo ☐ Mastered

A particle has initial velocity $(2\mathbf{i})$ m s^{-1} and constant acceleration $(-3\mathbf{i} - 3\mathbf{j})$ m s^{-2}. Find its speed after 4 s and the angle its direction of motion makes with $\mathbf{i}$.

(4)

WORKING

Q5.2.25 **Constant acceleration: suvat** ●●○ **[7 marks]** ☐ Todo ☐ Mastered

A ball is projected vertically upwards at 11 m s^{-1} from a point 30 m above the ground.

(a) Find the greatest height above the ground. **(3)**

(b) Find the time taken to reach the ground. **(3)**

(c) State one modelling assumption. **(1)**

WORKING

Q5.2.26 **Connected particles and pulleys** ●●○ **[8 marks]** ☐ Todo ☐ Mastered

Particles A (1 kg) and B (7 kg) are connected by a light inextensible string over a smooth fixed pulley. The system is released from rest with B 2 m above the ground.

(a) Find the acceleration of the system and the tension in the string. **(5)**

(b) Find the speed with which B hits the ground. **(2)**

(c) State how you used the fact that the string is inextensible. **(1)**

WORKING

Q5.2.27 **Moments, equilibrium and centre of mass** ●●○ **[5 marks]** ☐ **Todo** ☐ **Mastered**

A uniform beam AB of length 4 m and weight 187 N rests horizontally on supports at A and B. A load of 85 N is placed 3.5 m from A. Find the reactions at the supports, and state one modelling assumption.

(5)

WORKING

Q5.2.28 **Newton's laws and resolving forces** ●●○ **[5 marks]** ☐ **Todo** ☐ **Mastered**

A particle of weight 18 N hangs in equilibrium from two light strings inclined at 45° and 45° to the horizontal. Find the tension in each string.

(5)

WORKING

Q5.2.29 Inclined planes ●●○ [6 marks] ☐ Todo ☐ Mastered

A particle of mass 9 kg slides down a rough plane inclined at α to the horizontal, where $\tan\alpha = \frac{5}{12}$. The coefficient of friction is 0.1.

Find the normal reaction and the acceleration of the particle.

(6)

WORKING

Q5.2.30 Variable acceleration with calculus ●●○ [5 marks] ☐ Todo ☐ Mastered

A particle moves in a straight line with velocity $v = 3t^2 - 8t + 7$ m s^{-1} at time t s.

(a) Find the acceleration when $t = 2$. **(2)**

(b) Find the displacement from the starting point when $t = 2$. **(3)**

WORKING

Q5.2.31 Projectiles ●●○ [7 marks] ☐ Todo ☐ Mastered

A particle is projected from a point O on horizontal ground with speed 10 m s^{-1} at angle α above the horizontal, where $\tan\alpha = \frac{3}{4}$.

Find (a) the time of flight, (b) the range, (c) the greatest height.

(a) **(3)**

(b) **(2)**

(c) **(2)**

WORKING

Q5.2.32 2D motion with vectors ●●○ [4 marks] ☐ Todo ☐ Mastered

A particle has initial velocity $(-3\mathbf{i} + 2\mathbf{j})$ m s^{-1} and constant acceleration $(1\mathbf{i} - 2\mathbf{j})$ m s^{-2}. Find its speed after 6 s and the angle its direction of motion makes with $\mathbf{i}$.

(4)

WORKING

Q5.2.33 Constant acceleration: suvat ●●○ [7 marks] ☐ Todo ☐ Mastered

A ball is projected vertically upwards at 18 m s^{-1} from a point 11 m above the ground.

(a) Find the greatest height above the ground. **(3)**

(b) Find the time taken to reach the ground. **(3)**

(c) State one modelling assumption. **(1)**

WORKING

MY NOTES — 5.2 MECHANICS: MISTAKES TO REMEMBER

Answers and mark schemes

M = method mark, A = accuracy mark (depends on the preceding M), B = independent mark, ft = follow-through, * = given answer. Numerical answers are to 3 s.f. unless stated.

5.1 Statistics

Q5.1.01 Histograms, box plots, cumulative frequency, outliers · 4 marks

(a) 0.500, 1.30, 1.60, 1.10, 0.433

M1 frequency ÷ class width · A1

(b) below -8.5 or above 91.5

M1 IQR $= 25$ · A1

Q5.1.02 Measures of location and spread, coding · 5 marks

(a) mean 37.35, s.d. 14.57

B1 mean · M1 $\sqrt{\frac{\Sigma x^2}{n} - \bar{x}^2}$ · A1

(b) mean 3.235, s.d. 1.457

B1ft · B1ft

Q5.1.03 The large data set · 2 marks

different hemisphere/season and climate; not representative of UK summer

B1 per valid point · B1 per valid point

Q5.1.04 Binomial distribution · 7 marks

(a) 0.2522

M1 ${}^{10}C_3(0.35)^3(0.65)^7$ · A1

(b) 0.9740

B1

(c) 0.4862

M1 $1 - \mathrm{P}(X \leq 3)$ · A1

(d) 0.4601

M1 $\mathrm{P}(X \leq 6) - \mathrm{P}(X \leq 3)$ · A1

Q5.1.05 Probability: Venn diagrams and set notation · 9 marks

(a) Art only 11, both 8, Biology only 5, neither 26

B1 both · B1 only regions · B1 neither

(b) $\frac{4}{25}$; $\frac{9}{10}$; $\frac{8}{13}$

B1 · M1A1 · B1

(c) $\mathrm{P}(A)\mathrm{P}(B) = \frac{247}{2500} \neq \frac{4}{25}$ so not independent

M1 · A1

Q5.1.06 Populations and sampling methods · 2 marks

advantage: no sampling frame needed, quick; disadvantage: not random, interviewer bias

B1 per valid point · B1 per valid point

Q5.1.07 Correlation and regression · 3 marks

(a) each extra unit of x changes ice creams sold by 0.31

B1

(b) 23.5; extrapolation, so unreliable

B1 · B1

Q5.1.08 Histograms, box plots, cumulative frequency, outliers · 4 marks

(a) 1.40, 1.10, 2.90, 0.550, 0.867

M1 frequency ÷ class width · A1

(b) below -4 or above 68

M1 IQR $= 18$ · A1

Q5.1.09 Measures of location and spread, coding · 5 marks

(a) mean 36.00, s.d. 16.21

B1 mean · M1 $\sqrt{\frac{\Sigma x^2}{n} - \bar{x}^2}$ · A1

(b) mean 2.800, s.d. 1.621

B1ft · B1ft

Q5.1.10 The large data set · 2 marks

trace: less than 0.05 mm; treated as 0

B1 per valid point · B1 per valid point

Q5.1.11 Binomial distribution · 7 marks

(a) 0.2402

M1 ${}^{14}C_3(0.25)^3(0.75)^{11}$ · A1

(b) 0.5213

B1

(c) 0.4787

M1 $1 - \mathrm{P}(X \leq 3)$ · A1

(d) 0.0000

M1 $\mathrm{P}(X \leq 3) - \mathrm{P}(X \leq 3)$ · A1

Q5.1.12 Probability: Venn diagrams and set notation · 9 marks

(a) Art only 13, both 7, Biology only 12, neither 28

B1 both · B1 only regions · B1 neither

(b) $\frac{7}{60}$; $\frac{4}{5}$; $\frac{7}{19}$

B1 · M1A1 · B1

(c) $\mathrm{P}(A)\mathrm{P}(B) = \frac{19}{180} \neq \frac{7}{60}$ so not independent

M1 · A1

Q5.1.13 Populations and sampling methods · 3 marks

number students 1 to 900 (sampling frame); use a random number generator to pick 40 distinct numbers; ignore repeats

B1 per valid point · B1 per valid point · B1 per valid point

Q5.1.14 Correlation and regression · 3 marks

(a) each extra unit of x changes ice creams sold by 3.92

B1

(b) 96.7; interpolation, so reliable

B1 · B1

Q5.1.15 Histograms, box plots, cumulative frequency, outliers · 4 marks

(a) 1.20, 0.800, 2.50, 0.650, 0.933

M1 frequency ÷ class width · A1

(b) below -2.5 or above 73.5

M1 IQR $= 19$ · A1

Q5.1.16 Measures of location and spread, coding · 5 marks

(a) mean 36.66, s.d. 15.20

B1 mean · M1 $\sqrt{\frac{\Sigma x^2}{n} - \bar{x}^2}$ · A1

(b) mean 5.331, s.d. 3.039

B1ft · B1ft

Q5.1.17 The large data set · 2 marks

data not recorded/available; remove those days before calculating statistics

B1 per valid point · B1 per valid point

Q5.1.18 Binomial distribution · 7 marks

(a) 0.0025

M1 ${}^{24}C_5(0.5)^5(0.5)^{19}$ · A1

(b) 0.9887

B1

(c) 0.9967

M1 $1 - \mathrm{P}(X \leq 5)$ · A1

(d) 0.9854

M1 $\mathrm{P}(X \leq 17) - \mathrm{P}(X \leq 5)$ · A1

Q5.1.19 Probability: Venn diagrams and set notation · 9 marks

(a) Art only 14, both 9, Biology only 13, neither 24

B1 both · B1 only regions · B1 neither

(b) $\frac{3}{20}$; $\frac{47}{60}$; $\frac{9}{22}$

B1 · M1A1 · B1

(c) $\mathrm{P}(A)\mathrm{P}(B) = \frac{253}{1800} \neq \frac{3}{20}$ so not independent

M1 · A1

Q5.1.20 Populations and sampling methods · 2 marks

census: every member, fully accurate; sample: subset, quicker/cheaper/less data

B1 per valid point · B1 per valid point

Q5.1.21 Correlation and regression · 3 marks

(a) each extra unit of x changes test score by 3.36

B1

(b) 90.4; extrapolation, so unreliable

B1 · B1

Q5.1.22 Normal approximation to binomial · 5 marks

(a) n is large and $np = 70.8$, $n(1-p) = 106$ are both greater than 5 (p not close to 0 or 1)

B1

(b) $Y \sim \mathrm{N}(70.80, 42.48)$, $\mathrm{P}(Y < 72.5) = 0.6029$ (exact 0.6047)

B1 mean and variance · M1 continuity correction · M1 standardises · A1

Q5.1.23 Hypothesis testing: critical regions and p-values · 5 marks

$\mathrm{H}_0 : p = 0.4$, $\mathrm{H}_1 : p < 0.4$; $\mathrm{P}(X \leq 13) = 0.2112 > 0.1$; do not reject H_0: insufficient evidence that the proportion has decreased

B1 both hypotheses in terms of p · M1 $\mathrm{B}(40, 0.4)$ used · A1 p-value · M1 compares with significance level · A1 conclusion in context

Q5.1.24 Conditional probability and tree diagrams · 6 marks

(a) $\frac{2}{5}$

M1 tree diagram / sum of two products · A1 one product · A1

(b) $\frac{2}{3}$

M1 $\frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$ · A1ft numerator · A1

Q5.1.25 Normal distribution · 3 marks

$a = 91.37$

M1 $z = 0.8416$ · M1 $\frac{a-88}{4} = z$ · A1

Q5.1.26 Normal approximation to binomial · 5 marks

(a) n is large and $np = 31.0$, $n(1-p) = 31.0$ are both greater than 5 (p not close to 0 or 1)

B1

(b) $Y \sim \mathrm{N}(31.00, 15.50)$, $\mathrm{P}(Y < 36.5) = 0.9188$ (exact 0.9191)

B1 mean and variance · M1 continuity correction · M1 standardises · A1

Q5.1.27 Hypothesis testing: critical regions and p-values · 5 marks

$\mathrm{H}_0 : p = 0.5$, $\mathrm{H}_1 : p < 0.5$; $\mathrm{P}(X \leq 5) = 0.0207 < 0.05$; reject H_0: evidence that the proportion has decreased

B1 both hypotheses in terms of p · M1 $\mathrm{B}(20, 0.5)$ used · A1 p-value · M1 compares with significance level · A1 conclusion in context

Q5.1.28 Conditional probability and tree diagrams · 6 marks

(a) $\frac{5}{12}$

M1 tree diagram / sum of two products · A1 one product · A1

(b) $\frac{16}{25}$

M1 $\frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$ · A1ft numerator · A1

Q5.1.29 Normal approximation to binomial · 5 marks

(a) n is large and $np = 35.8$, $n(1-p) = 29.2$ are both greater than 5 (p not close to 0 or 1)

B1

(b) $Y \sim \mathrm{N}(35.75, 16.09)$, $\mathrm{P}(Y < 31.5) = 0.1447$ (exact 0.1447)

B1 mean and variance · M1 continuity correction · M1 standardises · A1

Q5.1.30 Hypothesis testing: critical regions and p-values · 5 marks

$\mathrm{H}_0 : p = 0.2$, $\mathrm{H}_1 : p > 0.2$; $\mathrm{P}(X \geq 6) = 0.1958 > 0.1$; do not reject H_0: insufficient evidence that the proportion has increased

B1 both hypotheses in terms of p · M1 $\mathrm{B}(20, 0.2)$ used · A1 p-value · M1 compares with significance level · A1 conclusion in context

Q5.1.31 Conditional probability and tree diagrams · 6 marks

(a) $\frac{31}{100}$

M1 tree diagram / sum of two products · A1 one product · A1

(b) $\frac{24}{31}$

M1 $\frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$ · A1ft numerator · A1

Q5.1.32 Normal distribution · 6 marks

$\mu = 122.0$, $\sigma = 9.00$

B1 $z = -0.8416$ · B1 $z = 1.2816$ · M1 $\frac{114.4-\mu}{\sigma} = -0.8416$ · M1 second equation · M1 solves simultaneously · A1 both

Q5.1.33 Hypothesis testing: normal mean and correlation · 6 marks

$\bar{X} \sim \mathrm{N}(90, \frac{17^2}{36})$; $z = -1.094$, p-value 0.1370; do not reject H_0

B1 $\mathrm{H}_0 : \mu = 90$, $\mathrm{H}_1 : \mu < 90$ · M1 distribution of $\bar{X}$ · M1 standardises · A1 p-value or z vs critical value · M1 comparison · A1 conclusion in context

Q5.1.34 Hypothesis testing: normal mean and correlation · 6 marks

$\bar{X} \sim \mathrm{N}(54, \frac{15^2}{20})$; $z = -2.385$, p-value 0.0085; reject H_0

B1 $\mathrm{H}_0 : \mu = 54$, $\mathrm{H}_1 : \mu < 54$ · M1 distribution of $\bar{X}$ · M1 standardises · A1 p-value or z vs critical value · M1 comparison · A1 conclusion in context

Q5.1.35 Normal distribution · 6 marks

$\mu = 132.0$, $\sigma = 9.00$

B1 $z = -1.2816$ · B1 $z = 0.8416$ · M1 $\frac{120.5-\mu}{\sigma} = -1.2816$ · M1 second equation · M1 solves simultaneously · A1 both

Q5.1.36 **Hypothesis testing: normal mean and correlation · 6 marks**

$\bar{X} \sim \text{N}(125, \frac{9^2}{20})$; $z = 1.640$, p-value 0.0505; do not reject H_0

B1 $\text{H}_0 : \mu = 125$, $\text{H}_1 : \mu > 125$ · M1 distribution of $\bar{X}$ · M1 standardises · A1 p-value or z vs critical value · M1 comparison · A1 conclusion in context

5.2 Mechanics

Q5.2.01 **Friction · 4 marks**

$2.19\ \text{m s}^{-2}$

M1 $R = 8g$ · M1 $F = \mu R$ · M1 $P - F = ma$ · A1

Q5.2.02 **Quantities and units, modelling assumptions · 2 marks**

dimensions negligible; no rotation, mass at a point

B1 meaning · B1 effect

Q5.2.03 **Motion graphs · 5 marks**

(a) trapezium shape

B1 shape · B1 values

(b) 282 m

M1 area of trapezium · A1

(c) $2.40\ \text{m s}^{-2}$

B1

Q5.2.04 **Friction · 4 marks**

$19.7\ \text{m s}^{-2}$

M1 $R = 3g$ · M1 $F = \mu R$ · M1 $P - F = ma$ · A1

Q5.2.05 **Quantities and units, modelling assumptions · 2 marks**

does not bend

B1 meaning · B1 effect

Q5.2.06 **Motion graphs · 5 marks**

(a) trapezium shape

B1 shape · B1 values

(b) 270 m

M1 area of trapezium · A1

(c) $4.50\ \text{m s}^{-2}$

B1

Q5.2.07 **Friction · 4 marks**

$7.02\ \text{m s}^{-2}$

M1 $R = 5g$ · M1 $F = \mu R$ · M1 $P - F = ma$ · A1

Q5.2.08 **Quantities and units, modelling assumptions · 2 marks**

mass evenly distributed so weight acts at the midpoint

B1 meaning · B1 effect

Q5.2.09 **Motion graphs · 5 marks**

(a) trapezium shape

B1 shape · B1 values

(b) 495 m

M1 area of trapezium · A1

(c) $2.25\ \text{m s}^{-2}$

B1

Q5.2.10 **Connected particles and pulleys · 8 marks**

(a) $a = 4.90\ \text{m s}^{-2}$, $T = 29.4\ \text{N}$

M1 equation for B: $6g - T = 6a$ · M1 equation for A: $T - 2g = 2a$ · A1 both correct · A1 a · A1 T

(b) $6.26\ \text{m s}^{-1}$

M1 $v^2 = 2as$ · A1

(c) both particles have the same acceleration (magnitude)

B1

Q5.2.11 **Moments, equilibrium and centre of mass · 5 marks**

$R_A = 76.0\ \text{N}$, $R_B = 76.0\ \text{N}$

M1 moments about A · A1 · M1 resolves vertically · A1 · B1 uniform rod / load as particle

Q5.2.12 **Newton's laws and resolving forces · 5 marks**

$T_1 = 9.50\ \text{N}$ (at 30°), $T_2 = 16.5\ \text{N}$ (at 60°)

M1 resolves horizontally · M1 resolves vertically · A1 equations · M1 solves · A1 both

Q5.2.13 **Inclined planes · 6 marks**

$R = 47.0$ N, $a = 1.80$ m s^{-2}

B1 $\sin\alpha = \frac{7}{25}$, $\cos\alpha = \frac{24}{25}$ · M1 resolves perpendicular · A1 R · M1 $F = \mu R$ · M1 Newton II along the plane · A1 a

Q5.2.14 **Variable acceleration with calculus · 5 marks**

(a) -13 m s^{-2}

M1 differentiates · A1

(b) $-\frac{355}{6}$ m

M1 integrates · M1 limits / $c = 0$ · A1

Q5.2.15 **Projectiles · 7 marks**

(a) 1.81 s

B1 components 21.2308 and 8.84615 · M1 $0 = U\sin\alpha\, t - 4.9t^2$ · A1

(b) 38.3 m

M1 horizontal distance · A1

(c) 3.99 m

M1 $v^2 = u^2 - 2gs$ · A1

Q5.2.16 **2D motion with vectors · 4 marks**

$\mathbf{v} = -3\mathbf{i} - 5\mathbf{j}$; speed 5.83 m s^{-1}; angle 121.0° from $\mathbf{i}$

M1 $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ · A1 · M1 magnitude · A1 angle

Q5.2.17 **Constant acceleration: suvat · 7 marks**

(a) 41.5 m

M1 $v^2 = u^2 + 2as$ with $v = 0$ · A1 height above start · A1

(b) 4.44 s

M1 $-30 = 15t - 4.9t^2$ · M1 solves · A1

(c) particle; no air resistance; g constant

B1

Q5.2.18 **Connected particles and pulleys · 8 marks**

(a) $a = 2.45$ m s^{-2}, $T = 36.8$ N

M1 equation for B: $5g - T = 5a$ · M1 equation for A: $T - 3g = 3a$ · A1 both correct · A1 a · A1 T

(b) 3.83 m s^{-1}

M1 $v^2 = 2as$ · A1

(c) both particles have the same acceleration (magnitude)

B1

Q5.2.19 **Moments, equilibrium and centre of mass · 5 marks**

$R_A = 131$ N, $R_B = 89.7$ N

M1 moments about A · A1 · M1 resolves vertically · A1 · B1 uniform rod / load as particle

Q5.2.20 **Newton's laws and resolving forces · 5 marks**

$T_1 = 27.0$ N (at 40°), $T_2 = 32.2$ N (at 50°)

M1 resolves horizontally · M1 resolves vertically · A1 equations · M1 solves · A1 both

Q5.2.21 **Inclined planes · 6 marks**

$R = 45.2$ N, $a = 1.96$ m s^{-2}

B1 $\sin\alpha = \frac{5}{13}$, $\cos\alpha = \frac{12}{13}$ · M1 resolves perpendicular · A1 R · M1 $F = \mu R$ · M1 Newton II along the plane · A1 a

Q5.2.22 **Variable acceleration with calculus · 5 marks**

(a) -28 m s^{-2}

M1 differentiates · A1

(b) -57 m

M1 integrates · M1 limits / $c = 0$ · A1

Q5.2.23 **Projectiles · 7 marks**

(a) 2.61 s

B1 components 9.6 and 12.8 · M1 $0 = U\sin\alpha\, t - 4.9t^2$ · A1

(b) 25.1 m

M1 horizontal distance · A1

(c) 8.36 m

M1 $v^2 = u^2 - 2gs$ · A1

Q5.2.24 **2D motion with vectors · 4 marks**

$\mathbf{v} = -10\mathbf{i} - 12\mathbf{j}$; speed 15.6 m s^{-1}; angle 129.8° from $\mathbf{i}$

M1 $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ · A1 · M1 magnitude · A1 angle

Q5.2.25 Constant acceleration: suvat · 7 marks

(a) 36.2 m

M1 $v^2 = u^2 + 2as$ with $v = 0$ · A1 height above start · A1

(b) 3.84 s

M1 $-30 = 11t - 4.9t^2$ · M1 solves · A1

(c) particle; no air resistance; g constant

B1

Q5.2.26 Connected particles and pulleys · 8 marks

(a) $a = 7.35$ m s^{-2}, $T = 17.2$ N

M1 equation for B: $7g - T = 7a$ · M1 equation for A: $T - 1g = 1a$ · A1 both correct · A1 a · A1 T

(b) 5.42 m s^{-1}

M1 $v^2 = 2as$ · A1

(c) both particles have the same acceleration (magnitude)

B1

Q5.2.27 Moments, equilibrium and centre of mass · 5 marks

$R_A = 104$ N, $R_B = 168$ N

M1 moments about A · A1 · M1 resolves vertically · A1 · B1 uniform rod / load as particle

Q5.2.28 Newton's laws and resolving forces · 5 marks

$T_1 = 12.7$ N (at 45°), $T_2 = 12.7$ N (at 45°)

M1 resolves horizontally · M1 resolves vertically · A1 equations · M1 solves · A1 both

Q5.2.29 Inclined planes · 6 marks

$R = 81.4$ N, $a = 2.86$ m s^{-2}

B1 $\sin\alpha = \frac{5}{13}$, $\cos\alpha = \frac{12}{13}$ · M1 resolves perpendicular · A1 R · M1 $F = \mu R$ · M1 Newton II along the plane · A1 a

Q5.2.30 Variable acceleration with calculus · 5 marks

(a) 4 m s^{-2}

M1 differentiates · A1

(b) 6 m

M1 integrates · M1 limits / $c = 0$ · A1

Q5.2.31 Projectiles · 7 marks

(a) 1.22 s

B1 components 8 and 6 · M1 $0 = U\sin\alpha\, t - 4.9t^2$ · A1

(b) 9.80 m

M1 horizontal distance · A1

(c) 1.84 m

M1 $v^2 = u^2 - 2gs$ · A1

Q5.2.32 2D motion with vectors · 4 marks

$\mathbf{v} = 3\mathbf{i} - 10\mathbf{j}$; speed 10.4 m s^{-1}; angle 73.3° from $\mathbf{i}$

M1 $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ · A1 · M1 magnitude · A1 angle

Q5.2.33 Constant acceleration: suvat · 7 marks

(a) 27.5 m

M1 $v^2 = u^2 + 2as$ with $v = 0$ · A1 height above start · A1

(b) 4.21 s

M1 $-11 = 18t - 4.9t^2$ · M1 solves · A1

(c) particle; no air resistance; g constant

B1